Introductory Calculus

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Direct substitution**

Find $\lim_{x\to 2}(3x^2 - 2x + 1)$.

$= 3(4) - 2(2) + 1 = 12 - 4 + 1 = 9.$

2 **[Foundational]** • **Target: Direct substitution**

Find $\lim_{x\to -1}(x^3 - 5x)$.

$= (-1)^3 - 5(-1) = -1 + 5 = 4.$

3 **[Foundational]** • **Target: Indeterminate forms: factor and cancel**

Find $\lim_{x\to 5}\frac{x^2-25}{x-5}$.

Substitution gives $\frac{0}{0}$. Factor: $\frac{x^2-25}{x-5} = \frac{(x-5)(x+5)}{x-5} = x + 5$ (for $x \neq 5$), so the limit is $5 + 5 = 10$.

4 **[Foundational]** • **Target: Indeterminate forms: factor and cancel**

Find $\lim_{x\to -3}\frac{x^2-9}{x+3}$.

Substitution gives $\frac{0}{0}$. Factor: $\frac{x^2-9}{x+3} = \frac{(x+3)(x-3)}{x+3} = x - 3$ (for $x \neq -3$), so the limit is $-3 - 3 = -6$.

5 **[Foundational]** • **Target: The derivative as a limit**

Write the difference-quotient definition of $f'(x)$.

$f'(x) = \lim_{h\to 0}\frac{f(x+h)-f(x)}{h}$.

6 **[Foundational]** • **Target: The derivative as a limit**

Using Theorem 5.1 ($f(x) = x^2 \Rightarrow f'(x) = 2x$), find the slope of the tangent line to $f(x) = x^2$ at $x = 3$.

$f'(3) = 2(3) = 6.$

7 **[Foundational]** • **Target: The derivative as a limit**

Using Theorem 5.1, find the slope of the tangent line to $f(x) = x^2$ at $x = -5$.

$f'(-5) = 2(-5) = -10.$

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 [Intermediate] • Target: Differential equations

Show that $y = x^2 - 7$ solves $\frac{dy}{dx} = 2x$.

By Theorem 5.1, the derivative of x^2 is $2x$, and the derivative of the constant -7 is 0. So $\frac{dy}{dx} = 2x + 0 = 2x$, matching the equation exactly.

9 [Intermediate] • Target: Differential equations

Explain why solving a differential equation gives a family of functions rather than a single number.

A differential equation only constrains the derivative, not the exact function; any constant added to a valid solution y has derivative 0 and doesn't change whether it solves the equation, so every $y =$ (that solution) $+C$ also works — a whole family, not one answer.

10 [Intermediate] • Target: Indeterminate forms: factor and cancel

Does $y = 5x^2 - 8$ solve $\frac{dy}{dx} = 10x$?

By Theorem 5.1, the derivative of a constant times x^2 scales by that constant, and a constant term's derivative is 0 — compare the result to the right-hand side of the equation. Final Answer: Yes — $\frac{dy}{dx} = 10x$, matching the equation.

11 [Intermediate] • Target: The derivative as a limit

Does $y = x^2 + 9$ solve $\frac{dy}{dx} = 2x$?

By Theorem 5.1, the derivative of a constant times x^2 scales by that constant, and a constant term's derivative is 0 — compare the result to the right-hand side of the equation. Final Answer: Yes — $\frac{dy}{dx} = 2x$, matching the equation.

12 [Intermediate] • Target: Differential equations

Find A^{-1} for $A = [3, 6;\ -4,\ -4]$.

$\det A = 12$. Swap the diagonal entries, negate the off-diagonal entries, divide by $\det A$: $A^{-1} = \left[-\frac{1}{3},\ -\frac{1}{2};\ \frac{1}{3},\ \frac{1}{4}\right]$.

13 [Intermediate] • Target: Limits & Continuity

Find A^{-1} for $A = [1, 3;\ -1, 5]$.

$\det A = 8$. Swap the diagonal entries, negate the off-diagonal entries, divide by $\det A$: $A^{-1} = \left[\frac{5}{8},\ -\frac{3}{8};\ \frac{1}{8},\ \frac{1}{8}\right]$.

14 [Intermediate] • Target: Derivatives & Power/Product Rules

Find $\lim_{x \to -2} \frac{x^2 - 2x - 8}{x + 2}$.

Final Answer: -6. Substitution gives $\frac{0}{0}$. Factor: $\frac{x^2 - 2x - 8}{x + 2} = \frac{(x + 2)(x - 4)}{x + 2} = x - 4$ (for $x \neq -2$), so the limit is $(-2) - 4 = -6$.

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Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Chain Rule & Extrema** • **Cross-Lesson: 4.01 Matrices & Vectors**

Find $\lim_{x \to -3} \frac{x^2 + x - 6}{x + 3}$.

Final Answer: -5. Substitution gives $\frac{0}{0}$. Factor: $\frac{x^2 + x - 6}{x + 3} = \frac{(x + 3)(x - 2)}{x + 3} = x - 2$ (for $x \neq -3$), so the limit is $(-3) - 2 = -5$.

16 **[Challenging]** • **Target: Definite Integrals & Fundamental Theorem** • **Cross-Lesson: 6.01 Probability**

Find $\lim_{x \to -3} \frac{x^2 + 9x + 18}{x + 3}$.

Final Answer: 3. Substitution gives $\frac{0}{0}$. Factor: $\frac{x^2 + 9x + 18}{x + 3} = \frac{(x + 3)(x + 6)}{x + 3} = x + 6$ (for $x \neq -3$), so the limit is $(-3) + 6 = 3$.

17 **[Challenging]** • **Target: Direct substitution** • **Cross-Lesson: 1.01 Complex Numbers**

Find $\lim_{x \to 1} \frac{x^2 + 6x - 7}{x - 1}$.

Final Answer: 8. Substitution gives $\frac{0}{0}$. Factor: $\frac{x^2 + 6x - 7}{x - 1} = \frac{(x - 1)(x + 7)}{x - 1} = x + 7$ (for $x \neq 1$), so the limit is $1 + 7 = 8$.

18 **[Challenging]** • **Target: Indeterminate forms: factor and cancel** • **Cross-Lesson: 2.01 Sequences & Series**

Find $\lim_{x \to 0} \frac{x^2 + 8x}{x}$.

Final Answer: 8. Substitution gives $\frac{0}{0}$. Factor: $\frac{x^2 + 8x}{x} = \frac{(x)(x + 8)}{x} = x + 8$ (for $x \neq 0$), so the limit is $0 + 8 = 8$.

19 **[Challenging]** • **Target: The derivative as a limit** • **Cross-Lesson: 3.01 Functions & Graphs**

Given $f(x) = -x^2 + 5x - 4$, use the power rule ($\frac{d}{dx}x^2 = 2x$, $\frac{d}{dx}x = 1$) to find $f'(2)$.

Final Answer: 1. $f'(x) = -2x + 5$, so $f'(2) = -2(2) + 5 = -4 + 5 = 1$.

20 **[Challenging]** • **Target: Differential equations** • **Cross-Lesson: 4.01 Matrices & Vectors**

Given $f(x) = 4x^2 + 5x + 9$, use the power rule ($\frac{d}{dx}x^2 = 2x$, $\frac{d}{dx}x = 1$) to find $f'(-6)$.

Final Answer: -43. $f'(x) = 8x + 5$, so $f'(-6) = 8(-6) + 5 = -48 + 5 = -43$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Direct substitution

Find $\lim_{x \to 0} (-2x^2 - 6x - 6)$.

a. -6 ✓

b. 6

c. -10

d. -2

$= -2(0)^2 - 6 = 0 - 6 = -6.$

22 [Foundational] • Target: Indeterminate forms: factor and cancel

Find $\lim_{x \to -5} (-3x^2 - 6x + 2)$.

a. -43 ✓

b. -39

c. -47

d. 43

$= -3(-5)^2 + 30 + 2 = -75 + 30 + 2 = -43.$

23 [Foundational] • Target: The derivative as a limit

Find $\lim_{x \to -3} (x^2 - 4x + 5)$.

a. 22

b. 26 ✓

c. -26

d. 30

$= (-3)^2 + 12 + 5 = 9 + 12 + 5 = 26.$

24 [Foundational] • Target: Differential equations

Find $\lim_{x \to -3} (2x^2 + 5x - 1)$.

a. -2

b. 6

c. 8

d. 2 ✓

$= 2(-3)^2 - 15 - 1 = 18 - 15 - 1 = 2.$

25 [Foundational] • Target: Limits & Continuity

Find $\lim_{x \to 4} (2x^2 + 5x + 9)$.

a. 57

b. -61

c. 65

d. 61 ✓

$= 2(4)^2 + 20 + 9 = 32 + 20 + 9 = 61.$

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26 [Foundational] • Target: Derivatives & Power/Product Rules

Find $\lim_{x\to -5}(-2x^2-8x+4)$.

a. 6

b. -10

c. -6 ✓

d. -2

$=-2(-5)^2+40+4=-50+40+4=-6.$

27 [Foundational] • Target: Chain Rule & Extrema

Find $\lim_{x\to -2}(-2x^2+5x-9)$.

a. 27

b. -23

c. -27 ✓

d. -31

$=-2(-2)^2-10-9=-8-10-9=-27.$

28 [Foundational] • Target: Definite Integrals & Fundamental Theorem

Find $\lim_{x\to -7}\frac{x^2+15x+56}{x+7}$.

a. 1 ✓

b. -2

c. -1

d. 4

Substitution gives $\frac{0}{0}$. Factor: $\frac{x^2+15x+56}{x+7}=\frac{(x+7)(x+8)}{x+7}=x+8$ (for $x\neq -7$), so the limit is $(-7)+8=1$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Direct substitution

Find $\lim_{x\to -5}\frac{x^2+6x+5}{x+5}$.

a. -1

b. -7

c. -4 ✓

d. 4

Substitution gives $\frac{0}{0}$. Factor: $\frac{x^2+6x+5}{x+5}=\frac{(x+5)(x+1)}{x+5}=x+1$ (for $x\neq -5$), so the limit is $(-5)+1=-4$.

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30 [Intermediate] • Target: Indeterminate forms: factor and cancel

Find $\lim_{x\to 5} \frac{x^2-11x+30}{x-5}$.

a. 1

b. -1 ✓

c. 2

d. -4

Substitution gives $\frac{0}{0}$. Factor: $\frac{x^2-11x+30}{x-5} = \frac{(x-5)(x-6)}{x-5} = x-6$ (for $x \neq 5$), so the limit is $5-6=-1$.

31 [Intermediate] • Target: The derivative as a limit

Find $\lim_{x\to 6} \frac{x^2-14x+48}{x-6}$.

a. -5

b. 2

c. -2 ✓

d. 1

Substitution gives $\frac{0}{0}$. Factor: $\frac{x^2-14x+48}{x-6} = \frac{(x-6)(x-8)}{x-6} = x-8$ (for $x \neq 6$), so the limit is $6-8=-2$.

32 [Intermediate] • Target: Differential equations

Find $\lim_{x\to 7} \frac{x^2-4x-21}{x-7}$.

a. 13

b. -10

c. 7

d. 10 ✓

Substitution gives $\frac{0}{0}$. Factor: $\frac{x^2-4x-21}{x-7} = \frac{(x-7)(x+3)}{x-7} = x+3$ (for $x \neq 7$), so the limit is $7+3=10$.

33 [Intermediate] • Target: Limits & Continuity

Find $\lim_{x\to -3} \frac{x^2+x-6}{x+3}$.

a. -2

b. -8

c. -5 ✓

d. 5

Substitution gives $\frac{0}{0}$. Factor: $\frac{x^2+x-6}{x+3} = \frac{(x+3)(x-2)}{x+3} = x-2$ (for $x \neq -3$), so the limit is $(-3)-2=-5$.

34 [Intermediate] • Target: Derivatives & Power/Product Rules

Find $\lim_{x\to 1} \frac{x^2+6x-7}{x-1}$.

a. 5

b. -8

c. 8 ✓

d. 11

Substitution gives $\frac{0}{0}$. Factor: $\frac{x^2+6x-7}{x-1} = \frac{(x-1)(x+7)}{x-1} = x+7$ (for $x \neq 1$), so the limit is $1+7=8$.

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35 [Intermediate] • Target: Chain Rule & Extrema

Given $f(x) = -3x^2 + 9x - 2$, use the power rule ($\frac{d}{dx}x^2 = 2x$, $\frac{d}{dx}x = 1$) to find $f'(3)$.

a. -7

b. -11

c. 9

d. -9 ✓

$f'(x) = -6x + 9$, so $f'(3) = -6(3) + 9 = -18 + 9 = -9$.

36 [Intermediate] • Target: Definite Integrals & Fundamental Theorem

Find $\lim_{x\to\infty} \frac{6x+3}{x-9}$.

a. $\frac{13}{2}$

b. -6

c. $\frac{11}{2}$

d. 6 ✓

Divide numerator and denominator by the highest power of x (here x): as $x \to \infty$, the constant terms vanish, leaving $\frac{6}{1} = 6$.

37 [Intermediate] • Target: Direct substitution

Find $\lim_{x\to\infty} \frac{x+5}{3x-8}$.

a. $\frac{5}{6}$

b. $\frac{1}{3}$ ✓

c. $-\frac{1}{3}$

d. $-\frac{1}{6}$

Divide numerator and denominator by the highest power of x (here x): as $x \to \infty$, the constant terms vanish, leaving $\frac{1}{3}$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Indeterminate forms: factor and cancel • Cross-Lesson: 2.01 Sequences & Series

Find $\lim_{x\to\infty} \frac{-4x-2}{-4x+4}$.

a. $\frac{3}{2}$

b. 1 ✓

c. $\frac{1}{2}$

d. -1

Divide numerator and denominator by the highest power of x (here x): as $x \to \infty$, the constant terms vanish, leaving $\frac{-4}{-4} = 1$.

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39 [Challenging] • Target: The derivative as a limit • Cross-Lesson: 3.01 Functions & Graphs

Find the average rate of change of $f(x) = x^2$ on the interval from $x = 3$ to $x = 1$.

a. 2

b. 6

c. -4

d. 4 ✓

$\frac{f(1) - f(3)}{1 - 3} = \frac{1^2 - 3^2}{-2} = \frac{-8}{-2} = 4$.

40 [Challenging] • Target: Differential equations • Cross-Lesson: 4.01 Matrices & Vectors

Find the average rate of change of $f(x) = x^2$ on the interval from $x = 3$ to $x = 4$.

a. 5

b. -7

c. 9

d. 7 ✓

$\frac{f(4) - f(3)}{4 - 3} = \frac{4^2 - 3^2}{1} = \frac{7}{1} = 7$.

41 [Challenging] • Target: Limits & Continuity • Cross-Lesson: 6.01 Probability

Evaluate $\lim_{x \to 0} \frac{\tan(3x)}{x}$.

a. 3 ✓

b. 1

c. $\frac{1}{3}$

d. 0

$\lim_{x \to 0} \frac{\sin(3x)}{x\cos(3x)} = 3\lim_{x \to 0} \frac{\sin(3x)}{3x} \cdot \frac{1}{\cos(3x)} = 3(1)(1) = 3$.

42 [Challenging] • Target: Derivatives & Power/Product Rules • Cross-Lesson: 1.01 Complex Numbers

Find the derivative of $f(x) = 4x^3 - 5x^2 + 7x - 9$.

a. $12x^2 - 10x + 7$ ✓

b. $12x^2 - 5x + 7$

c. $4x^2 - 10x + 7$

d. $12x^3 - 10x^2 + 7$

By power rule: $\frac{d}{dx}(4x^3) = 12x^2$, $\frac{d}{dx}(-5x^2) = -10x$, $\frac{d}{dx}(7x) = 7$.

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43 [Challenging] • Target: Derivatives & Power/Product Rules • Cross-Lesson: 2.01 Sequences & Series

What is the derivative of $g(x) = \frac{2x+1}{x-3}$?

a. $-\frac{7}{(x-3)^2}$ ✓

b. $\frac{7}{(x-3)^2}$

c. $\frac{2}{(x-3)^2}$

d. $-\frac{5}{(x-3)^2}$

Quotient rule: $\frac{2(x-3)-(2x+1)(1)}{(x-3)^2} = \frac{2x-6-2x-1}{(x-3)^2} = -\frac{7}{(x-3)^2}$.

44 [Challenging] • Target: Chain Rule & Extrema • Cross-Lesson: 3.01 Functions & Graphs

At what value of x does $f(x) = 2x^3 - 9x^2 + 12x + 1$ have a local minimum?

a. $x = 2$ ✓

b. $x = 1$

c. $x = 0$

d. $x = 3$

$f'(x) = 6x^2 - 18x + 12 = 6(x^2 - 3x + 2) = 6(x-1)(x-2) = 0$. $f''(x) = 12x - 18$. At $x = 2$: $f''(2) = 6 > 0$ (local min).

45 [Challenging] • Target: Definite Integrals & Fundamental Theorem • Cross-Lesson: 4.01 Matrices & Vectors

Evaluate the definite integral: $\int_0^3 (3x^2 - 2x + 1)dx$.

a. 21 ✓

b. 18

c. 24

d. 27

Antiderivative is $[x^3 - x^2 + x]_0^3 = (27 - 9 + 3) - 0 = 21$.

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